

Madness in the Probe

The Master Reference for the Probe Sub-Programme of the Genesis Information Project

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How to read this

This is the master reference for the probe sub-programme: the exploration of the probe of Axiom 5 — its versions, its fusion, its completeness, its costs, its nesting — and of **Madness**, the proposed object marking where the probe’s questions fail to receive lawful values from the shadow. It is a **spoke** of *The Genesis Information Project* (the main hub, whose axioms, conventions, and honesty boundary it inherits unamended) and a **hub** for the companion notes and experiments this sub-programme will produce. In its initial versions it carries more detail than a mature hub should; detail migrates to spokes as they are written, and this document will thin to grades and pointers.

Provenance. The originating ideas are Alexander Pisani’s (*Madness_In_the_Probe.md*, July 2026). A first-pass corpus mapping was contributed by a separate Claude instance (*first-pass notes*, July 2026). An exploratory sweep with pre-committed measures (x520 scripts: the rad-projection ledger, the extraction-rate grade, the fusion algebra) was run by the lead instance in the same cycle; its results are provisional until formally rerun and banked, and are marked (*previewed*) below. **Change log.** v0.2 (same cycle, after the second sweep): the disagreement register resolved at the originator’s direction (§8); E8 added to the register (§7); §10 extended. v0.3 (E1 return): §3.1 banked; E1 register entry closed; lead verification addendum appended to the E1 results spoke. v0.4 (E4 return): §5 M_1 banked; E4 register entry closed with the M6 design flaw owned; §10 self-priced thread updated and its census closed; lead verification addendum appended to the E4 results spoke. v0.5 (E2 return): §3.4 banked with the conservation-frontier resolution; E2 register entry closed; lead verification addendum appended to the E2 results spoke. v0.6 (E3 return): §5 M_2 banked with the blindness ledger, the energy map, the D2 theorem, and M4b; E3 register entry closed; lead verification addendum appended to the E3 results spoke. v0.7 (distillation, by sign-off): the M_2 three-state taxonomy (Blind/Wrong/Fused) adopted; the GenIS-side component of M_2 closed; M_2 re-scoped as repertoire-relative Madness on the probe’s side, its Tarski-shaped border with M_4 noted; the §6 predicted-degeneracy observation recorded as consistent-with, not confirmation-of. v0.8 (E5 return): P1’s mechanics banked into §1 (conserved budget, the unremovable bit, poles-only-by-assertion, the zero-surcharge theorem, the P2 rhyme); E5 register entry closed; lead verification addendum appended to the E5 results spoke. v0.9 (E7 return): §4 banked (insight-as-conversion, the ceiling and detector, the sufficiency summit, foreclosure and the safe floor, the max-ent refinement); E7 register entry closed; lead verification addendum appended to the E7 results spoke.

Conventions. Fact / Reading / Lead, in the senses fixed by the main spine; measures committed before code runs; promote/park/kill resolved against pre-registered criteria; proto/shadow hygiene where genesis-layer terms appear.

The metaphysics rule (recorded decision, this cycle). This sub-programme deliberately works ground the main corpus fenced off. The rule adopted: metaphysical framings are admitted **during exploration**, each marked (*conceptual scaffolding*), because they are part of how the originating intuitions are held and tested. At **distillation**, every scaffolding item is either formalised into a graded claim or an owned posit, or retired into the record the way the architectural metaphor was retired — preserved, credited, and no longer load-bearing. The final work states only what is derived from, or honestly posited alongside, the axioms. Nothing load-bearing may ever rest on an owned posit alone.

On the name. *Madness* is retained as the term of art (recorded decision), in the tradition of *imaginary* and *chaos*: the word carries the intended meaning — a break the probe cannot rationalise — better than a neutral coinage would, and it names a family (the object M , the types M_1 – M_4 , the Mad kernel, the Mad projection). If the programme’s tests kill the useful content, or the word proves overbearing at distillation, it is dropped then, not before.

1. The Posit Set (owned, caged)

Four posits are adopted at the standing of the main spine’s Axioms 4–5 and the Erasure Axiom: owned, not derived; load-bearing only jointly with graded results; each unfalsifiable by construction where marked, and stated so once.

P1 — Virtual–Virtual Dualism. *The observer is a virtual projection that cannot certify the system projecting it; existence, for the observer, is unresolved between self and system.* The **primary formal statement** is Axiom 3’s mechanics applied to the self/system question: realness is a maximum-entropy distribution over the generator-superposition, pruned by probe returns and re-spread over the survivors. The two classical positions are the degenerate point masses — materialism (System = 1, Self = 0) and solipsism (Self = 1, System = 0) — and the interior values the frame wants are not stipulated: they fall out of maximum entropy, since no lawful sequence of returns discriminates the poles (*Reading; E5 tests the mechanics*). The scalar **Realness budget**, Self + System = 1, is retained as the two-outcome marginal of that distribution — a guide, not the headline (*recorded decision; original formulation preserved in §8*). The two poles of the originating document map to the corpus’s own poles — lattice bottom 1 ($\ln 1 = 0$, the empty message) and top 0 (infinite resistance) — with “energy as displacement across the poles” reading as resistance itself, and the far pole read only through erasure, which is the Erasure Axiom verbatim (Paper 9 §§2–3). “The system could be timeless; the probe exists between snapshots” is the Regime-1/Regime-2 split applied to time (Note B §8; the succession-channel note). (*Mappings: Reading.*) **Mechanics banked (E5, 530_e5_realness.py, July 2026 — 19/19 committed checks; independently rerun, identical; the fence maintained: none of this bears on P1’s truth).** Under the rendered barrier: the budget’s interior values are **conserved, not stipulated** — frozen at max-ent $(\frac{1}{2}, \frac{1}{2})$ by every lawful act, recomputed never enforced; lawful evidence learns everything *except one bit* — the posterior converges to the true twin pair and terminates at $H = \ln 2$ exactly, **the unremovable existence bit**; the poles (materialism, solipsism) are reachable **only by the probe asserting its own status** — every within-algebra conditioning leaves the marginal untouched, so the entropy-deficit road and the self-assertion road coincide, and the asserted pole is **absorbing**

(Cromwell); the leak control confirms the freeze is the barrier's doing (at $\varepsilon = 0.05$, lawful evidence reaches the pole from either direction). Adopted from the run's report-only observations, verified as an elementary theorem: **when the barrier holds, the wrong-pole dogmatist's predictive distribution is identical to the open probe's — being wrong about the existence bit costs exactly zero observationally, becoming E2's surcharge (divergently) only as the barrier leaks.** Dogmatism about the existence bit is free exactly when the bit is unobservable: the observational face of P1, and the measured form of “the diagonal has no purchase.” The structural rhyme is banked alongside: the frozen marginal is to P1 what the rotation gauge is to P2 — the two places in the corpus where evidence terminates one step short of a determination. (*Invariance, consistency, absorption, the identical-predictives theorem: Fact, elementary/classical. The Realness readings, the existence bit, poles-only-by-assertion, the rhyme: Reading.*)

P2 — Qualia incommunicability. *The kernel of an observation — what a Mad object is like — cannot be communicated between probes.* This is philosophy's hard problem and is carried as an **owned posit**, not a result. What distinguishes its standing here is that it arrives with a **proven structural shadow**: the *Natural Key* note established that the geometry of the resistance space — every distance and overlap, which is everything the shareable quantitative channel carries — is invariant under rotation of the basis, while only discreteness (succession) pins the basis. Two probes can therefore agree on all shareable structure while their kernel bases differ by a rotation: *which axis is red* is gauge. The same theorem explains both halves — quantities communicate because succession's discreteness forces a shared lattice; kernels do not because nothing on the Mad side pins the rotation. The inverted-spectrum scenario is a gauge transformation. (*Rotation invariance: Fact, prior corpus. The qualia identification: Reading. The incommunicability itself: posit.*)

P3 — Mad systems. *Systems that project without following logic may exist.* Unfalsifiable by construction, exactly as the core genesis posit is; held at the same owned standing; **nothing downstream depends on it**, and this clause is part of the posit.

P4 — The host probe. *The probe cannot disprove that its record store is a subset of a larger probe's records — the vat argument run upward.* (The originating image: a dreaming alien holding knowledge the dream-probe cannot access.) Unfalsifiable by construction; owned. Its only permitted downstream content is the **nesting mechanism of §4, which stands on Axiom 5 alone and does not require P4** — the posit adds the upward direction, never machinery.

2. Definitions

Madness, M (provisional definition). Madness is the failure of a probe's question to receive a lawful value from the shadow: structure or queries for which the genesis information system, read through erasure, cannot return an answer the probe can rationalise. The definition is deliberately held provisional — the typology of §5 is mapped first, and whether one functional unifies it is a pre-registered test (§6, E6), not an assumption. Before the genesis operation, 0 and 1 are not yet distinct; the originating intuition — *Madness as the forced absence of NOT* — is given a precise face in the Mad projection below.

Mad kernel $\times$ quantitative multiplicities (the decomposition). Every observation factors as a **Mad kernel** — *which* irreducible generators are present: idempotent (repetition adds nothing, $\text{Blue}^2 = \text{Blue}$), incommunicable-in-itself (P2), gauge under rotation — times **quantitative multiplicities** — *how many* of each: non-idempotent, shareable, pinned to the lattice by dis-

creteness. “Seven black dots” = kernel {black, dot} $\times$ multiplicity 7. The arithmetic model is the radical decomposition: $n = \text{rad}(n) \times \text{multiplicities}$. The idempotence hinge underneath is *measured*, not assumed — Note A’s F1 established that idempotent structure carries no power ladder and non-idempotent structure does. (*Decomposition as the formalisation of the qualia/quantity split: Reading, the sub-programme’s strongest candidate; F1 as its measured base: Fact, prior corpus.*)

The Mad projection Q and the idempotence toll (*previewed, x520*). Define $Q(n) = \text{rad}(n)$: forget multiplicity, keep the kernel. Three results:

1. Q is idempotent and a **join-morphism**: $\text{rad}(ab) = \text{lcm}(\text{rad } a, \text{rad } b)$ — verified exactly for all $a, b \leq 300$. Q converts the non-idempotent multiplicative merge into the idempotent join: $\text{Blue}^2 = \text{Blue}$ as an equation. (*Fact, elementary.*)
2. Q is not new machinery: it is the main spine’s **intuitionistic double negation** (§2.1 — Heyting $\neg\neg$ collapses exponents to present-or-absent, the squarefree skeleton). So the Mad kernel is *what survives the lossy NOT-NOT*, and quantity is *exactly what the classical, involutive NOT preserves and the lossy one destroys*: the originating “forced absence of NOT” lands on the corpus’s existing two-NOTs distinction. (*Fact, restatement of prior corpus; the Madness identification: Reading.*)
3. The erasure price of Q — the **idempotence toll** — is $T_Q(n) = \ln(n/\text{rad } n)$, and its mean over the integers is a universal constant:

$$\bar{T}_Q = \sum_p \frac{\ln p}{p(p-1)} \approx 0.7554 \text{ nats} \approx 1.09 \text{ bits}$$

(measured 0.7538 at $N = 10^6$; the gap is the slow tail). Forgetting quantity costs about one bit on average. The constant has a zeta identity: $\bar{T}_Q = \sum_{p, k \geq 2} \Lambda(p^k) p^{-k}$ — the $\beta = 1$ mass of the innovation spectrum’s **graded middle**. Equivalently, the Zeta Circuit’s mean energy splits as

$$-\frac{\zeta'}{\zeta}(\beta) = \underbrace{\sum_p \ln p \, p^{-\beta}}_{\text{kernel } (k=1)} + \underbrace{\sum_p \frac{\ln p}{p^\beta(p^\beta - 1)}}_{\text{ladder } (k \geq 2)},$$

verified with residuals matching the PNT tail estimates of the truncated sums to three digits; the divergence as $\beta \rightarrow 1^+$ is carried **entirely by the kernel**, while the quantitative ladder contributes the finite constant $\bar{T}_Q$ at $\beta = 1$. (*All identities elementary/classical-adjacent — $\ln(n/\text{rad } n)$ and the constant are standard analytic-number-theory material, and no new theorem is claimed; the assembly, the toll reading, and the tie to the Madness split are the contribution.*)

Consciousness-event (scoped definition; deflationary; exhaustive). *A consciousness-event is a diagonal-query event: an act in which a probe set puts to its system a question whose evaluation requires reading the probe’s own complement.* Nothing else in this corpus is meant by “consciousness.” No claim about experience, wakefulness, moral status, or the common usage of the word is made or implied, and the definition deliberately does not coincide with any of them; it exists so that the theorems that govern diagonal queries (Gödel, Tarski — §5, M_4) apply verbatim to the thing named. The originating narrower form — a probe querying the system about its own existence — is the paradigm instance. (*Definition; adopted with the firm deflation clause by recorded decision.*)

Insight (deflationary definition; see §4). *The conversion of record-virtual structure into record-actual structure within a probe’s own store: computation on the probe’s side of the barrier, priced on the probe’s own ledger, involving no outer query.*

3. Grounded Probe Theory

The parts of the originating document that already have exact corpus homes. These are the sub-programme’s floor.

3.1 Knowledge states and fusion (*banked: E1, 525_e1_fusion.py, July 2026*). A probe’s knowledge is an exponent vector over channels — the same object as an integer’s prime-exponent vector. Then combined knowledge = coordinatewise max = lcm; shared knowledge = coordinatewise min = gcd; and Paper 1’s union law reads, word for word, as inclusion–exclusion of information:

$$\Omega_{\text{lcm}} = \Omega_a + \Omega_b - \Omega_{\text{gcd}}, \quad I(\text{probe}_1; \text{probe}_2) = \ln \text{gcd}.$$

Fusion is attained per channel by one contributor — “a system of probes is only as good as its best expert, channel by channel” — and the attainment lifts to k -ary fusions ($k = 2$ –6, E1’s completion). Banked exact on both sides: 2×10^4 random vector pairs to 8.5×10^{-14} ; exhaustive integers $a, b \leq 200$ to 1.8×10^{-15} , with the min gcd, max lcm reconstruction structurally exact throughout. The full lattice laws hold (idempotence, commutativity, associativity, absorption — vector side and independent integer side), carrying E1’s flag: **fusion is a join, so pooled knowledge is kernel-like, not accumulation-like** — a probe fused with itself learns nothing; pooling is not learning. This is the M_3 kernel theme surfacing on the probe side: joins collapse repetition, exactly as Q does. (*Algebra and lattice laws: Fact, classical/prior corpus. Probe interpretation and the idempotence flag: Reading. Lead verification: independent rerun exact; two definitionally-vacuous checks noted in the results spoke’s addendum, conclusions unaffected.*)

3.2 What crosses between probes, and what cannot. Only the lattice-pinned quantitative content is shareable (multiplicities, distances, correlations); the kernel is rotation-gauge (P2’s shadow). Communication is bounded by the lesser repertoire unless teaching — repertoire transfer — occurs, which is itself priced (the repertoire is a codebook; its cost is its description length — the *Probe* note’s rent).

3.3 Completeness is an asymptote. A complete probe (all channels = 1) is not distant but impossible: the innovator stream is inexhaustible from inside (Note A F1 addendum; main spine §2.4, Pillar 2 — Gödel through the universality identification). The learning trajectory is real; the terminus is not — a strengthening, since the framework *predicts* endless learning. Two of the originating document’s sentences are kept as its most defensible: a probe **cannot certify its own completeness on a channel from inside** (that certification is a read of its own complement — Tarski-shaped), and probes **can infer their own incompleteness but not the other’s completeness**. The deficiency vector $cA-A, cB-B, \dots$ is the *Two Ignorances* note’s second ignorance, coordinatised. (*Classical pillars: Fact. Probe frame: Reading.*)

3.4 The probe’s energy (*banked: E2, 521_e2_ledger.py, July 2026 — 22/22 committed checks; independently rerun by the lead, log-identical*). Learning is conversion of virtual to actual (Axiom 5’s final clause), and compression’s discards are erasures **on the probe’s own ledger** — now measured. The originating question — *balanced, or net stored?* — resolves as a **conservation frontier**: ingested information splits with no residue between paid erasure and retained storage at every archive fraction f (storage = fH , payment = $(1-f)H$, sum = ingested exactly), with the poles the **summarizer** ($f=0$: pays H forever, stores nothing) and the **hoarder** ($f=1$: all-reversible, pays nothing, memory grows at H). Storage is exactly memory growth; payment is exactly memory bound. What knowledge changes is the *rate*, not the law: **a learned model is a**

discount on the price of forgetting worth exactly the captured redundancy ($\ln 4 - H = 0.1733$ measured as 0.1719); **model error is a surcharge of exactly** $KL(p||q)$ (0.4332 measured as 0.4326) — so a confident wrong model costs *more than ignorance* whenever KL exceeds the redundancy, as here; and **learning’s permanent residue grows as the MDL term** $\frac{k-1}{2} \ln n$ (slope -1.5038 against -1.5) — what survives forgetting the details is the model, logarithmically. Bonus banked as the implementer’s A1: total forgetting cost is **route-independent per realization** (forget-order-then-summary = discard-raw, to 10^{-14}) while the payment’s *location* differs — the first live instance of E8’s state-vs-process split inside another experiment. (*Identities: Fact, classical — Landauer, Bennett, Shannon incl. the mismatched-code identity, Rissanen; the frontier resolution, discount/surcharge/residue readings, and the ledger-as-model-audit: Reading. The mandatory caveat travels: simulation confirms the accounting’s consistency; the carrier reading stays a Lead pending physics.*)

3.5 The Bayes seam. The pruning mechanics of Axiom 5 (constrain, never disclose; re-spread at maximum entropy) is the Bayesian face of the frame; its geometry-of-incomplete-knowledge connection (Paper 7; the Fisher digest) is a marked thread, not yet worked.

4. The Nested Tower (*new this cycle*)

Axiom 5 is self-applicable, and the originating “dreaming alien” splits cleanly under it. An observer is a probe set *holding records*; the record store is itself an information system; by Axiom 5 it carries its own native probe set — the observer’s **inner probe**. Structure the records conserve but the inner repertoire cannot read is **virtual to the observer, inside the observer**. Then: **insight** is the inner conversion (definition, §2) — the blank “filled in” was already conserved in the records, and the felt suddenness is the inner repertoire acquiring the read; the energy of insight is the erasure cost of the inner computation, “background” only in that it is spent on the probe’s own ledger rather than at the outer interface; **dreaming** is inner-probe activity with the outer interface closed (*conceptual scaffolding*); and Axiom 3 run inwardly — maximum entropy over record-consistent completions — is the formal face of filling in the blanks. One hard constraint comes free and is adopted as E7’s pre-registered prediction: by the **data-processing inequality, insight can never increase information about the generator** — inner conversion, never transfer. What the originating image called knowledge passing from the alien to the dreamer is, inside the frame, always and only the inner repertoire expanding over what the records already held. The upward direction — that our records are themselves a subset of a host probe’s — is P4, and stays a posit. (*Self-application of Axiom 5: Reading. Insight-as-inner-conversion: definition. The DPI bound: Fact, classical.*)

Mechanics banked (E7, 535_e7_nested.py, July 2026 — **33/33 committed checks, fully exact, no Monte Carlo; independently rerun by the lead, identical; both fences maintained: no claim about human cognition, no bearing on P4**). The tower’s definitions are now priced. **Insight is real conversion**: the inner probe, on a closed record store, gained 0.259 nats about the generator with zero outer queries — purely by upgrading its reader — and the **record ceiling held against the full battery**: no coarsening, composition, or independently-randomized post-processing exceeded its DPI bound, with independent noise sitting *exactly on* the ceiling (the equality case) while the **leaky oracle — an outer channel disguised as inner noise — was caught 0.202 nats above it**. The banked criterion: *within-ceiling gains are always conversion; above-ceiling gains are always communication* — P4’s boundary made operational, P4

itself untouched. **The ladder saturates at the minimal sufficient statistic, not at the raw record** (Fisher–Neyman, measured to 3×10^{-16} across independent partitions): the records conserve 2.237 nats of structure with *exactly zero* generator-content — “what the records conserve” and “what they say about the world” exhibited as different sets, the finding in one number being the $S_2 \rightarrow S_3$ ratio: **2.237 nats of inner erasure actualising zero θ -information. Foreclosure and its safe floor:** discard below sufficiency at a closed interface and the insight is unreachable forever — no re-purchase exists, the closed-interface hardening of E3’s path (iii) — while discarding *down to* sufficiency loses nothing: **compression is insight-safe exactly down to minimal sufficiency, and forecloses below it** — E2’s hoarder–summarizer frontier acquires its named safe floor. One refinement to this section’s own earlier sentence: **max-ent-inwardly is conditioning exactly at the sufficient rung and is biased below it** — the fill-in error at S_1 (0.029) is the unread structure itself, a non-sufficiency signature. (*DPI, sufficiency, chain rule: Fact, classical, attributed. Insight-as-conversion priced, the ceiling as insight-capacity and conversion/communication criterion, the saturation summit, foreclosure and insight-safe compression, the max-ent refinement: Reading, E7’s.*)

5. The Madness Typology (graded map)

Four types were proposed; each is mapped, graded, and given its test. Whether they are four things or one is §6’s question, not assumed either way.

M_1 — **unbounded rationalisation cost** (banked: *E4, 534_e4_rates.py, July 2026 — 17/17 committed checks; independently rerun by the lead, identical*). The grade: $M_1\text{-rate}(x) = \lim_k \frac{1}{k} \ln q_k$ over continued-fraction convergents — nats about x extracted per question asked. The banked structure:

- **The floor is exact at every finite k , not just asymptotically:** partial quotients ≥ 1 force $q_k \geq F_{k+1}$, so $\text{rate}_k(x) \geq \text{rate}_k(\varphi)$ for every x at every k — banked with minimum margin exactly 0, attained by φ alone, across 100 random reals and every named object. The $1/k$ drift fit recovered slope -0.32351 against the predicted $\ln \varphi - \ln \sqrt{5} = -0.32351$, intercept $= \ln \varphi$. The golden ratio is the maximal-Madness point of Type 1 **by theorem, now exact rather than conjectural**. (*Fact, classical — Hurwitz/Fibonacci; the exactness-at-finite- k sharpening banked here.*)
- **The floor is self-referential:** every metallic ratio $[a; a, a, \dots]$ is a fixed point, $\text{rate}(x) = \ln x$ — learned at exactly its own resistance rate — banked for $a = 1\text{--}5$ with shrinking deviations, and φ is the minimal fixed point. (*Fact, elementary; the Mad-pole reading: Reading.*)
- **The rate reads the tail, not the value** — banked in its sharpest possible form: $\sqrt{2}$, the silver ratio $1+\sqrt{2}$, and metallic $a=2$ returned **byte-identical** $\text{rate}_{200} = 0.880582$, because they present the probe with the same questions; different-sized numbers, one rate. Note B’s “the value is the shadow; the path is the process,” surfacing inside the grade itself. (*Fact.*)
- **Typical reals sit at Lévy’s constant** $\pi^2/(12 \ln 2) = 1.18657$ (banked: mean 1.18193, sd 0.0724, $n = 100$ at $k = 200$; the small gap is the finite- k correction), and **e is anti-Mad:** its rate climbs $0.9931 \rightarrow 1.1868 \rightarrow 1.3938 \rightarrow 1.5617$ across checkpoints, unbounded — more learnable per question than a typical real. (*Fact, classical — Lévy; Euler’s continued fraction of e .*)

Indeterminate forms, the type’s other exemplar, already carry their residue: a contest between two infinite resistances that only rates can settle (Note B §2). Type 1 is *finite information forever*

arriving at the slowest lawful rate — resistance without terminus, not lawlessness. (*Reading.*)

M_2 — **NOT suspension** (banked: *E3*, `524_e3_monotone.py`, July 2026 — 28/28 committed checks, fully exact, no Monte Carlo; independently rerun by the lead, identical). Removing NOT is the **monotone fragment**, whose expressible events are the up-sets; the genesis probe (the meet-atom) is *already* monotone, negation a derived complement-read. The standing prediction — *suspension buys blindness, not access* — banked in quantified form, with the originator’s extension delivered as a measured map:

- **The blindness is total, not expensive:** on the 4-prime cap-3 lattice, every complement-shaped feature (primality, squarefreeness, $e_2=0$) has decision-blindness $B = 1$ **exactly** — no up-set beats the trivial guess (max-weight closure/min-cut, integer-exact) — while the up-event control $\omega \geq 2$ is expressed at $B = 0$ exactly. The certifiability table lands the r.e./co-r.e. split on the lattice: squarefreeness refutable-never-verifiable (240/256 vs 0), $\omega \geq 2$ the mirror (243/256 vs 0), and $n = 1$ **monotone-invisible** — all silence, nothing verifiable or refutable, the maximal residual of any state. Suspension’s blindness is worst exactly where the classical answer would have been “no.”
- **The energy map:** the monotone probe pays 2.3671 of the lattice’s $\ln 256 = 5.5452$ nats and leaves $\ln 24 = 3.1781$ — 57.3%, **exactly** $\ln 24 / \ln 256$ — **permanently unread as unremoved superposition**. The withheld NOT’s cost is not stored work and there is nothing to reinject: suspension declines to buy pruning; what it “saves” is what it never learns.
- **The D2 theorem** (the register’s first theorem born from a resolved disagreement): *deferral is free* — suspend-then-lift costs exactly pay-now, to 0.00, a route-equality of independently simulated ledgers — but *fusing under suspension is priced*: erase a channel’s record (believe “5=2”) and completing knowledge later costs an excess of exactly $\ln 4 + (\ln 4 - \frac{1}{4} \ln 24) = 1.9781$ nats — the erasure paid to forget plus the pruning re-purchased. Belief is free; the count is not.
- **M4b, the record-vs-knowledge identity:** the monotone probe’s record physically pins the channel ($H = \ln 4$) while its certifiable knowledge is only the pruning paid (0.5918); the uncertifiable surplus **equals the residual blindness exactly** (0.7945). A probe cannot certify its own completeness from inside (§3.3), now priced on the ledger.

The originating observation retained as the type’s seed — *before concepts are identified as NOT one another, they are fused* — and the fusion hierarchy (Q fuses repetitions; total NOT-removal fuses kernels, the primordial indeterminate of Note B §7) remains the parked upper storey. (*Closure/min-cut, Post’s split, Landauer/Bennett: Fact, classical, attributed; Razborov’s walls cited as context. The blindness ledger, the unremoved-superposition finding, the D2 theorem, and the priced completeness-uncertifiability: Reading, E3’s.*)

Distillation (adopted by sign-off, July 2026, post-E2/E3). The GenIS-side component of M_2 is closed: there is no system-side break — the monotone world is one-sided but perfectly lawful, and an identification like “2=5” never reaches the system at all; it is a **filing policy on the probe’s side**. On that side the ledger distinguishes three states the original framing compressed into one: **Blind** (the distinction never resolved — zero paid, superposition held, no debt; deferral free; ordinary ignorance, not Madness), **Wrong** (the identification *used* as a model — a per-use surcharge of exactly KL , E2’s mismatched eraser; mispricing, not Madness), and **Fused** (the distinguishing record destroyed — erasure paid plus the re-purchase debt when the count is demanded, E3’s 1.9781; an irreversible act with a computed unwind cost, not Madness). Belief is free until use converts holding into counting. The paired statements banked as the type’s summary: **NOT**

converts silence into information, and the monotone probe is incapable of error and therefore incapable of deciding — E2’s *being wrong costs more than knowing nothing* and E3’s *never being wrong means never knowing* are the two jaws of one vice, with commitment the motion between them. What survives as M_2 proper is the residue none of the three states touches: the part **unclosable from inside the repertoire** — silence never converts, $n=1$ is invisible forever, completeness is uncertifiable (M4b). M_2 is hereby re-scoped as **repertoire-relative Madness on the probe’s side**: questions unposable within the fragment, not unanswered by the system. This residue is Tarski-shaped, so M_2 ’s deepest layer borders M_4 — the convergence pattern again. Marked observation, consistent-with not confirmation-of: the Blind state’s signature (0 paid / 0 removed) is the 0/0 degeneracy §6’s candidate functional assigned to Type 2 before E3 was designed; E6 remains the kill test. (*Taxonomy and re-scoping: Reading, adopted; the ledger signatures underneath: Fact, banked.*)

M_3 — **Mad objects (qualia)**. The mixing mathematics has a classical home: **Grassmann’s laws** (1853) — perceptual colour is a convex cone; every colour splits into chromaticity (direction — idempotent) $\times$ intensity (magnitude — additive), which is the kernel $\times$ multiplicity decomposition again, in its native habitat. The pink formula, in both registers at once: the originating self-correction $\text{pink} = \text{white} \times \text{red}$ (kernel union with red-weighted multiplicities) and the cone form $\text{pink} \propto (1+w)R + wY + wB$ are the same statement — desaturation adds the complement pair; nothing is subtracted. Incommunicability is P2 (posit, with the rotation-gauge shadow as its proven part). The formal-vs-framework axiom placement across the barrier is retained as a tidy Reading whose cash value is that only the formal side has a measurable innovation profile. (*Grassmann: Fact, classical. Decomposition: Reading. Incommunicability: posit.*)

M_4 — **the self-referencing probe**. The diagonal, already resident: no probe set reads its own complement (Tarski); self-referential queries are Gödel’s diagonal (Note A §6). The default position (recorded decision; original preserved in §8): the system does not answer “no” — the query is **unanswerable from inside**, carrying no assignable value; nothing is wiped out; the question has no purchase. Grading: $M_4(q)$ = the degree to which evaluating q requires the asker’s own complement, with the **Turing jump hierarchy** supplying rungs — each jump one level of self-reference resolved by a strictly stronger probe. “Primeness as a probe-learned feature” is already a corpus principle (primality as output; the probe’s concept of 5 bundles quantity with irreversibility because the probe built the concept from erasure records — *Natural Key, Primality Shadow*). Consciousness-events (§2’s definition) are exactly the M_4 -maximal queries. (*Classical theorems: Fact. Grading scaffold: Reading. Definition as recorded.*)

6. The Candidate Unification (Lead, held at arm’s length)

The candidate. One functional behind the typology:

$$M(q) = \frac{\text{erasure paid}}{\text{generator-superposition entropy removed}} \quad \text{— the price per nat of pruning.}$$

The four types would then be its four **degeneracies**: $M_1 = \infty/\infty$, settled only by rates (l’Hôpital reading the ledger — Note B verbatim, with φ the extremal rate); $M_3 =$ positive price over zero return *in the shareable currency* (the return is pure rotation-gauge); $M_2 = 0/0$ (refuse the NOT, pay nothing, learn nothing — blindness with a form); M_4 not indeterminate but **undefined** — the diagonal admits no lawful evaluation, off the pole arithmetic entirely. If this holds, Madness is

one quantity and the typology is its degeneracy structure, vindicating the originating instinct that indeterminate forms sit at the centre.

The flag, recorded deliberately. This is the sweep’s highest-excitement, largest-knowledge-gap item: the assignments above are pattern-matched, not derived — precisely the condition under which the excitement heuristic warns that work gets pushed off course. It is therefore held as a **Lead with a pre-registered kill test (E6)** and adopted nowhere else in this document: define M concretely on the Stage-1 machines plus a monotone-restricted probe plus a diagonal query; **kill** if the four types fail to land on distinct degeneracies without hand-tuning; **park** if some land and others need machinery the frame lacks; **promote** only if all four land and the assignment survives an adversarial re-labelling check.

7. The Experiment Register (pre-registered)

Adopted and designed by the lead instance (recorded decision), with E1–E5 carried from the first-pass notes where agreed, E3 extended, and E6–E7 added this cycle. Priority order as listed. Environmental protocol: PID-unique filenames throughout.

- **E1 — Probe-fusion algebra. BANKED (July 2026; 525_e1_fusion.py, seed 20260710; independently rerun by the lead, residuals identical).** All pre-registered measures exact; kill condition never approached. §3.1 promoted. E1’s additions beyond the preview: the k -ary attainment completion ($k = 2$ –6) and the lattice-law/idempotence flag. Results spoke: *E1_Fusion_Algebra_Results.md* (with lead verification addendum).
- **E4 — The Type-1 grade. BANKED (July 2026; 534_e4_rates.py, seed 20260710; independently rerun by the lead, identical; 17/17 committed checks).** Floor exact at finite k with margin 0 attained by φ ; metallic fixed points $a = 1$ –5; the tail control at its sharpest (byte-identical rates for $\sqrt{2}$ /silver/metallic-2); Lévy typicality $n = 100$; e anti-Mad. The committed M6 census was sign-infeasible on the implementer’s legitimate $(0, 1)$ sampling — **a design flaw in the lead’s brief, owned in the results spoke’s addendum**; the implementer’s labelled M6b addition supplies the thread’s data point, and the census question is closed theoretically in §10. Results spoke: *E4_Extraction_Rate_Results.md* (with lead verification addendum).
- **E2 — The probe’s own ledger. BANKED (July 2026; 521_e2_ledger.py, seed 20260710; independently rerun by the lead, log-identical; 22/22 committed checks).** The balanced-or-stored question resolved as the conservation frontier (§3.4); the knowledge discount, the KL surcharge, and the MDL residue all measured on the pinned closed forms; pre-run pins C1–C4 honoured exactly; three labelled implementer additions including A1 (per-realization route-equality of total forgetting cost). Results spoke: *E2_Probe_Ledger_Results.md* (with lead verification addendum).
- **E3 — The monotone probe (extended). BANKED (July 2026; 524_e3_monotone.py; fully exact, 28/28 committed checks; independently rerun by the lead, identical).** The prediction confirmed in quantified form: total decision-blindness ($B = 1$ exactly) on every complement-shaped feature; $\ln 24$ nats (57.3%) permanently unread as unremoved superposition — nothing stored, nothing reinjectable; the D2 theorem (deferral free to 0.00; fusion-under-suspension excess exactly 1.9781 nats); M4b’s priced completeness-uncertifiability (lead amendment, pre-run). Pre-run clarifications C1–C4 honoured; C1 corrected a lead display-artifact caught by the §5.5 derivation check. Results spoke:

E3_Monotone_Probe_Results.md (with lead verification addendum).

- **E5 — Realness by maximum entropy. BANKED (July 2026; 530_e5_realness.py, seed 20260710; independently rerun by the lead, identical; 19/19 committed checks).** P1’s mechanics coherent, strengthened from “stays interior” to *conserved*: the marginal frozen at $(\frac{1}{2}, \frac{1}{2})$; terminal entropy exactly $\ln 2$ (the unremovable bit); the coincidence claim confirmed operationally (poles only by self-assertion, absorbing); the leak dichotomy carried the non-vacuity; the strike-at-P1 condition never triggered. The zero-surcharge theorem adopted into P1’s block. Fence maintained throughout: no bearing on P1’s truth. Results spoke: *E5_Realness_Results.md* (with lead verification addendum).
- **E7 — The nested probe and the insight ledger. BANKED (July 2026; 535_e7_nested.py; fully exact, 33/33 committed checks; independently rerun by the lead, identical).** Insight priced as conversion (0.259 nats at zero outer queries); the ceiling held against the full DPI battery with the leaky oracle caught 0.202 above it (the conversion/communication criterion); the ladder saturates at minimal sufficiency (S-sufficiency collapse caught by the implementer at the §5.5 check — the lead’s design error, corrected pre-run; the corrected finding stronger than the original bar); foreclosure below sufficiency, insight-safe compression at it (M4b, lead amendment). Results spoke: *E7_Nested_Probe_Results.md* (with lead verification addendum).
- **E8 — The two-path merge (state vs process; the interference question).** The multiplicative twin of the *Additive Merge* note’s banked path-function result, plus that note’s parked phase Lead, adopted here. **Stage A (classical ledger):** assemble a fixed n from its prime multiset by distinct merge trees under the two-legged ledger. Committed: the arithmetic value is the **state function** — exact $\ln n$ on every path (the isomorphism as consistency check); the storage/dissipation split is the **path function** (scout, this cycle: the per-merge overlap sum $\sum \ln \gcd(\text{left}, \text{right})$ is already path-dependent — p^4 : balanced $4 \ln p$ vs chain $3 \ln p$ — semantics to be pinned to the ledger note’s conventions at design time); characterise the extremal trees, with the pre-registered structural claim that the extremal shape is **Huffman-like and kernel-blind** (a function of the exponent vector only, never of which primes); measure how many bits of the tree survive in the record, and verify that erasing the record recovers exactly the multiplicative shadow and nothing else. **Stage B (contingent):** couple the succession-derived proto-phase (*Phase from the Probe*; *Phase on the Merge*) to merge legs and ask whether two paths to the same n can **cancel** — a null at fixed n that no mixture of classical ledgers produces. Promote if cancellation is exhibited without importing quantity; park Stage B if proto-phase cannot couple lawfully; kill the interference reading if apparent cancellation reduces to classical bookkeeping.
- **E6 — The Madness functional (kill test of §6).** Last, deliberately: it needs E3’s machinery and E2/E4’s vocabulary. Criteria as stated in §6.

8. Open Disagreements (live register)

The register stands open by recorded decision: entries are added when a default and an originating formulation part ways, and are resolved here rather than silently overwritten.

D2 — Suspended NOT as a tool against M_1/M_4 : resolved to the default, by the originator, this cycle. The closing rationale is recorded because it sharpens E3’s frame: **suspension is non-conservative** — a result obtained under a suspended NOT cannot be re-embedded in the full logic without reopening the problem it “solved,” so agreement under suspension is a fact about

the audience’s belief-state, not a link between the objects. E3 proceeds unchanged: the question is no longer whether suspension solves, but what the blindness costs and where the withheld energy sits.

Three further entries (the wipe-out narrative, its answered-no variant, and the budget-as-headline) were withdrawn or superseded at the originator’s direction this cycle and are removed; their content is not carried.

9. Boundary Discipline (stated once)

Nothing in this sub-programme is, or can become, evidence that the genesis system holds beyond the axioms that posit it; hosting a structure is not evidence, provability is not truth, and no leverage is claimed on any open problem. The posits P1–P4 are unfalsifiable by construction and are marked so here, once. The classical mathematics is not ours — Gödel, Tarski, Turing, Chaitin, Landauer, Bennett, Shannon, Jaynes, Hurwitz, Lévy, Grassmann, Razborov, and the rest; the contribution is the axioms’ extension into the probe’s interior, the mappings, the graded readings, and a small number of new formal observations, each marked at its point of appearance.

10. Open Threads

- **The validation target (origin vs notation), recorded at the originator’s direction this cycle.** The origin posit is unfalsifiable by construction and stays so; validation therefore cannot mean proof. The standing target is **surplus structure**: a quantitative relation among probe-side observables that is *forced* if the genesis origin is taken seriously and *coincidental* if the framework is mere notation — the shape of Bell’s move, where a metaphysical-sounding posit became load-bearing because the formalism built on it produced a checkable inequality its rivals had no reason to expect. Independent convergences (the corpus’s strongest evidence class) are the informal version; the target is to make one convergence quantitative and forced. Current best candidate: E8 Stage B.
- **The Madness operations inventory** (recorded intent): Q (the Mad projection — defined, priced), the suspension operator (E3 — to be constructed), the M functional (E6 — candidate), the inner-conversion operator of insight (E7 — to be constructed). Whether these compose — e.g. whether suspension is Q ’s completion from multiplicity-fusion to kernel-fusion — is open.
- **The idempotence toll’s variance and higher cumulants:** $\bar{T}_Q$ is the mean; the full distribution of $\ln(n/\text{rad } n)$, and whether its cumulants live on the graded middle the way the corpus’s do, is unworked.
- **The Bayes/Paper 7 seam** (§3.5): the geometry of the deficiency vector; Fisher information of the pruning process.
- **Repertoire transfer (teaching):** fusion (E1) is per-channel max of *existing* knowledge; the dynamics of one probe raising another’s channel — priced as rent transfer — is unmapped.
- **The fusion hierarchy of M_2 :** multiplicity-fusion (Q) $\rightarrow$ kernel-fusion (primordial indeterminate); whether intermediate fusions exist and what they cost.

- **The fixed-point class of the M_1 grade** (“self-priced numbers”): $\{x : \text{rate}(x) = \ln x\}$ contains every metallic ratio (banked, E4). The random-real census is now **closed, theoretically and observationally**: by Lévy’s theorem almost every real has the *same* rate (Lévy’s constant), so a typical real is self-priced only at the single point $x = e^{\pi^2/(12 \ln 2)} = 3.2758\dots$ — the Khinchin–Lévy constant, pleasingly the a.s. limit of $q_k^{1/k}$ itself — making the class measure-zero among typicals; E4’s labelled M6b census found 0/200 hits (nearest miss 1.6×10^{-3}) against a computed expectation of ≈ 0.05 hits, the null being the $\approx 95\%$ -likely outcome. The thread’s remaining content is the **atypical-rate solutions**: beyond the metallic family (period-1), do other quadratics solve $\text{rate}(x) = \ln x$ — e.g. period-2 $[a; b, a, b, \dots]$, where the condition becomes an eigenvalue equation on the period matrix? Cheap to explore.
- **Time in the tower**: the inner probe’s snapshots vs the outer interface’s clock; connects to the main spine’s marked time hole.

11. Document Role (recorded decision)

Hub and spokes, inherited. This document is the sub-programme’s hub — posits, definitions, the graded map, the experiment register, the open-disagreements register — and a spoke of *The Genesis Information Project*. Companion notes to come are its spokes; the hub grades and points, and thins as they are written. The five axioms of the main spine are not amended here; any candidate for adoption into the main spine (none yet) would be proposed there explicitly.

The originating axioms, hypotheses, and directional intuitions — Madness and its typology, Virtual–Virtual Dualism, the probe programme, the nested-probe direction — are Alexander Pisani’s. The formalisation, the corpus mappings (including a first-pass mapping by a separate Claude instance), the exploratory computations, the experiment designs, and this compilation were developed jointly with Claude (Anthropic). Adoption and amendment recorded jointly, July 2026. Living document; rebuilt as the sub-programme is better understood.